

Chapter 2

Example Problems Related to Soil Identification and Classification

2.0 GENERAL COMMENTS

One of the foremost aims in soil mechanics has been to find methods for discriminating between different kinds of soil in a given category. The properties on which these distinctions are based are known as *index properties*. The tests required to determine the index properties are referred to as *classification tests*. When discussing soil grain properties it is convenient to divide soil into *cohesionless* (gravel, sand, silt) and *cohesive* (clay) soils.

The nature of any soil can be altered by appropriate manipulation. For example, vibrations transform a loose sand into a dense one. The behavior of a soil thus depends not only on the significant properties of the *individual* constituents of the soil mass, but also on those properties that are due to the *arrangement* of the particles within the mass. It is thus convenient to divide index properties into two classes:

- *Soil aggregate properties*. This involves analyzing a soil mass in its intact state (as much as possible). The stress history is thus taken into consideration. The most significant aggregate property of cohesionless soils is the *relative density*, whereas that of cohesive soils is the *consistency*.
- *Soil grain (or individual) properties*. The soil characteristics are based on the *size*, *shape*, and *distribution* of the particles. In clay soils, the mineralogical character of the smallest grains is of importance. No consideration of the stress history is included in the determination of such properties.

2.1 PARTICLE SIZES

Perhaps the most fundamental soil grain property is the size of a particle. A single linear dimension cannot uniquely define the size of a particle, other than a cube or a sphere. Thus, particle (grain) size typically refers to the *diameter* of a soil grain. The meaning of “particle size” therefore depends on the

dimension that was recorded and how it was obtained. The sizes of soil particles vary over a very large range.

Depending on the *predominant* size of particles within a soil, it is generally referred to as being a *gravel*, *sand*, *silt*, or *clay*.

Gravels are unconsolidated rock fragments produced by mechanical weathering with occasional particles of quartz, feldspar, and other minerals.

Sand particles are produced by mechanical weathering and are made of mostly quartz and feldspar, though other mineral grains may also be present.

Silts are the microscopic soil fractions that consist of very fine quartz grains and some flake (plate)-shaped particles that are fragments of micaceous minerals; they are produced by chemical and physical weathering.

Clays are mostly flake (plate)-shaped microscopic and submicroscopic particles of mica, clay minerals, and other minerals.

- The clay minerals are products of chemical weathering of feldspars, ferromagnesians, and micas.
- The clay minerals give the plastic (putty like) property to soils.
- Clays also exhibit *cohesiveness*; i.e., the ability to stick together. As a result, such soils are thus referred to as being *cohesive*; by contrast, soils consisting of gravels, sands, and silts are referred to as *cohesionless*.
- The presence of *water* greatly affects the engineering response of clays.
- The most important clay minerals are kaolinite, illite, and montmorillonite.

Since they are larger in size, gravels and sands are referred to as being *coarse grained*. By contrast, silts and clays are referred to as being *fine grained*. A conventional dividing line between these two groups is the approximate smallest particle sizes that are visible to the naked eye, typically 0.060 or 0.075 mm.

Coarse-grained soils are thus *nonplastic* and *noncohesive* (or cohesionless). Among fine-grained soils, silts are generally *nonplastic* and *noncohesive*, while clays are both *plastic* and *cohesive*.

To describe soils by their particle size, several organizations have developed particle size classifications. Since there is no clear-cut division between the four general soil groups (i.e., gravel, sand, silt, and clay), and since particle sizes vary widely, several classifications exist, with different points of delimitation. Some examples of such classifications are given below.

Remark: The range of particle sizes from 200 to 0.002 mm spans *five* (5) orders of magnitude!

Table 2.1 lists the grain size scale proposed by the Swedish chemist and agricultural scientist Albert Atterberg (1846–1916).

Table 2.2 lists the grain size scale proposed by the US Department of Agriculture (USDA).

TABLE 2.1 Grain Size Ranges According to A. Atterberg

Name of Soil	Size Range (mm)
Boulders	>200
Cobble (stone)	60–200
Coarse gravel (pebble)	20–60
Medium gravel	6–20
Fine gravel	2–6
Coarse sand	0.6–2
Medium sand	0.2–0.6
Fine sand	0.06–0.2
Coarse silt	0.02–0.06
Medium silt	0.006–0.02
Fine silt	0.002–0.006
Clay	<0.002

TABLE 2.2 Grain Size Ranges According to the US Department of Agriculture

Name of Soil	Size Range (mm)
Very coarse sand	1.00–2.00
Coarse sand	0.50–1.00
Medium sand	0.25–0.50
Fine sand	0.10–0.25
Very fine sand	0.05–0.10
Silt	0.002–0.05
Clay	<0.002

Finally, [Table 2.3](#) lists the grain size scale proposed by the Unified Soil Classification System (USCS).

Remark: Those particles classified as being “clay” on the basis of their size may not necessarily contain clay minerals.

TABLE 2.3 Grain Size Ranges According to the Unified Soil Classification System	
Name of Soil	Size Range (mm)
Boulders	>300
Cobbles	75–300
Gravel	4.75–75
Sand	0.075–4.75
Fines (silt and clay)	<0.075

Remark: Nonclay soils typically contain quartz, feldspar, or mica particles that are small enough to be less than 0.002 mm (2 μ m) in size.

Remark: It is thus appropriate for particles smaller than 0.002 or 0.005 mm to be called “clay-size” particles rather than clay.

2.2 DISTRIBUTION OF GRAIN SIZES

In classifying a soil it is important to determine the size range of particles present in a soil (typically expressed as a percentage of the total dry weight). Two approaches are generally used to determine the soil particle size distribution. A *sieve analysis* is used for particle sizes greater than or equal to 0.075 mm. For particle sizes smaller than 0.075 mm, a *hydrometer analysis* is used.

Remark: Both tests significantly disturb soil samples from their in situ state.

2.2.1 Sieve Analysis

A sieve analysis involves the shaking of a soil sample through a set of sieves that has progressively smaller square openings. The results of a sieve analysis are plotted on a particle size distribution or *gradation* curve. In such a curve, the particle size (in mm) is plotted as the abscissa on a logarithmic scale. The ordinate on such curves is the percent passing a specific sieve size.

The classification of the particle size distribution of a soil is referred to as *soil gradation*. Coarse-grained soils, mainly gravels or sands, are graded as either *well graded* or *poorly graded*.

Well-graded soils contain particles having a wide range of sizes. They have a good representation of all sizes from No. 4 sieve (4.75 mm) to No. 200 sieve (0.075 mm).

Poorly graded soils do *not* have a good representation of all sizes of particles from No. 4 to No. 200 sieve. Poorly graded soils are further divided into *uniformly graded* or *gap-graded* soils.

A *uniformly graded* soil is a soil that has most of its particles at about the same size. An example of a uniformly graded soil is one in which only sand of the No. 20 sieve size (0.850 mm) is present.

A *gap-graded* soil is a soil that has an excess or deficiency of certain particle sizes or a soil that has at least one particle size *missing*. An example of a gap-graded soil is one in which sand of the No. 10 (2.00 mm) and No. 40 sizes (0.425 mm) are missing, and all the other sizes are present.

2.2.2 Quantities Computed From Gradation Curves

To facilitate the interpretation of gradation curves, let D_{10} , D_{30} , and D_{60} equal to the grain diameter (in mm) corresponding to 10%, 30%, and 60%, respectively, passing by weight (or mass).

The *coefficient of uniformity* C_u , which is a crude shape parameter, is calculated using the following equation:

$$C_u = \frac{D_{60}}{D_{10}} \quad (2.1)$$

If $C_u = 1.0$, the soil contains primarily *one* grain size. The smaller the C_u , the *more uniform* will be the associated gradation.

The *coefficient of curvature* C_c , which is a shape parameter, is calculated using the following equation:

$$C_c = \frac{(D_{30})^2}{D_{10}D_{60}} \quad (2.2)$$

Once the coefficient of uniformity and the coefficient of curvature have been calculated, they must be compared to published gradation criteria. For example, both C_u and C_c are used in the Unified Soil Classification System (USCS).¹ In particular, for a gravel to be classified as being *well graded*

$$C_u > 4 \quad \text{and} \quad 1 < C_c < 3 \quad (2.3)$$

If both of these conditions are not met, the gravel is classified as being *poorly graded*. Section 2.5 gives additional details pertaining to the USCS.

If both of these criteria are met, the gravel is classified as *well graded*. If, however, both of these criteria are not met, the gravel is classified as being *poorly graded* (GP).

1. Section 2.5 discusses the Unified Soil Classification System in greater detail.

For a sand to be classified as *well graded*, the following criteria must be met:

$$C_u > 6 \quad \text{and} \quad 1 < C_c < 3 \quad (2.4)$$

If both of these criteria are met, the sand is classified as *well graded*. If both of these criteria are not met, the sand is classified as being *poorly graded*.

2.2.3 Importance of Soil Gradation

Soil gradation is very important to geotechnical engineering. It is an indicator of other engineering properties such as *compressibility*, *shear strength*, and *hydraulic conductivity*.

In a design, the gradation of the in situ or on site soil often controls the design and ground water drainage of the site. A poorly graded soil will have better drainage than a well-graded soil because there are more void spaces in a poorly graded soil.

When a fill material is being selected for a project such as a highway embankment or earthen dam, the soil gradation is considered. A well-graded soil is able to be compacted more than a poorly graded soil. These types of projects may also have gradation requirements that must be met before the soil to be used is accepted.

When options for ground remediation techniques are being selected, the soil gradation is often a controlling factor.

2.2.4 Hydrometer Analysis

A hydrometer analysis attempts to determine the size distribution of fine-grained soils. It is based on the principle of sedimentation of soil particles settling in water. Hydrometer analyses are typically performed on particles passing the No. 200 sieve (0.074 mm). The data from a hydrometer analysis allow for the determination of the suspension density.

In determining the distribution of particle sizes, a deflocculating agent is first applied to the soil, which has been mixed with water. On mixing, the particles are separated out and form a solution. When a soil sample is dispersed in water, the particles will settle at different velocities, depending on their size, shape, weight, and on the viscosity of water.

All particles are *assumed* to be spherical (Figure 2.1).

For a spherical particle of radius r , the upward buoyancy force is equal to the volume of fluid displaced multiplied by the unit weight of water; i.e.,

$$F_b = \frac{4}{3} \pi r^3 \gamma_w \quad (2.5)$$

The downward force is due to the weight of the particle; i.e.,

$$F_d = \frac{4}{3} \pi r^3 \gamma_s \quad (2.6)$$

where, as before, γ_s is the unit weight of the solid phase.

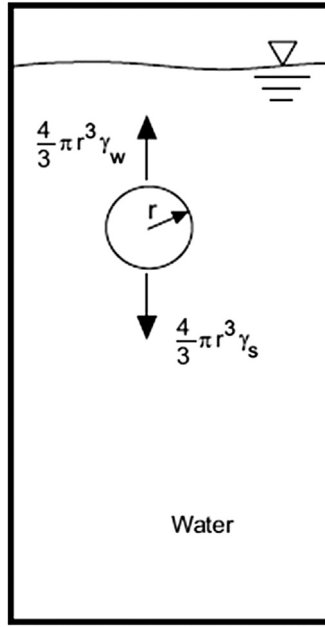


FIGURE 2.1 Schematic illustration of a single particle in suspension.

Stokes found that the net vertical force was related to the velocity of the particle according to the following relation:

$$\frac{4}{3} \pi r^3 \gamma_s - \frac{4}{3} \pi r^3 \gamma_w = 6\pi\eta r v \quad (2.7)$$

where η is the viscosity of water and v is the velocity of the sphere as it settles in the water. Solving for the velocity and relating this to the particle diameter D gives

$$v = \frac{2}{9} \frac{r^2}{\eta} (\gamma_s - \gamma_w) = \frac{2}{9} \frac{(D/2)^2}{\eta} (\gamma_s - \gamma_w) = \frac{(\gamma_s - \gamma_w) D^2}{18\eta} \quad (2.8)$$

Solving for this diameter gives

$$D = \sqrt{\frac{18\eta v}{(\gamma_s - \gamma_w)}} = \sqrt{\frac{18\eta}{(\gamma_s - \gamma_w)}} \sqrt{\frac{L}{t}} \quad (2.9)$$

Since $G_s = \gamma_s / \gamma_w \Rightarrow \gamma_s = G_s \gamma_w$, the above expression can also be written as

$$D = \sqrt{\frac{18\eta}{(G_s - 1)\gamma_w}} \sqrt{\frac{L}{t}} \quad (2.10)$$

At a depth L (commonly called the effective length) at time t , all particles have a diameter smaller than D , that is, all larger particles would have settled

beyond the zone of measurement. Hydrometers are thus designed to allow the amount of soil that is still in suspension to be determined.

In typical calculations associated with a hydrometer analysis, L is measured in centimeters (cm), the units of η are (g s)/cm², $\gamma_w = 1.0$ g/cm³, t is measured in *minutes*, and it is advantageous to represent D in units of *millimeters*. The above expression for D is thus modified so as to account for the aforementioned units, giving

$$D = \sqrt{\frac{18\eta}{(G_s - 1)\gamma_w}} \sqrt{\frac{L}{t} \left(\frac{\text{min}}{60 \text{ s}}\right)} \left(\frac{10 \text{ mm}}{\text{cm}}\right) = \sqrt{\frac{30\eta}{(G_s - 1)}} \sqrt{\frac{L}{t}} \quad (2.11)$$

The above expression is often written as

$$D = K \sqrt{\frac{L}{t}} \quad (2.12)$$

where D has units of *millimeters*, L has units of *centimeters*, t is expressed in *minutes*, and

$$K = \sqrt{\frac{30\eta}{(G_s - 1)}} \quad (2.13)$$

Since the viscosity η is a function of temperature, it follows that K will likewise be a function of temperature.

The standard conditions for hydrometer analyses (ASTM 152-H type hydrometer) are $G_s = 2.65$ at 20°C inside a 1000 ml fluid (usually distilled water) where no dispersing agent is added. If the hydrometer analysis is performed under conditions that differ from these calibration conditions, corrections need to be made. The four commonly made corrections are thus

1. *Temperature correction* (F_T): when the temperature differs from 20°C, the following correction must be made:²

$$F_T = -4.85 + 0.25T \quad (2.14)$$

where the test temperature T is between 15 and 28°C.

2. *Specific gravity correction* (a): when the specific gravity of the solids is not 2.65, the following correction must be made:

$$a = \frac{1.65G_s}{2.65(G_s - 1)} \quad (2.15)$$

3. *Zero correction* (F_z): if a dispersing (or deflocculating) agent is added to the soil–distilled water suspension, the zero reading from the hydrometer

2. Das, B.M., 2013. Soil Mechanics Laboratory Manual, eighth edition. Oxford University Press, New York.

will be changed. Thus, a correction to the deviation of reading from the zero mark of the hydrometer is required.

4. *Meniscus correction* (F_m): accounts for the reading error introduced when readings are taken off the upper level of the meniscus formed, at the stem of the hydrometer, by the soil–water suspension.

The effective length (L) must be related to the hydrometer reading (R), which ranges between 0 and 60 g/L. The magnitude of L is often computed from the following expression:

$$L = L_1 + \frac{1}{2} \left(L_2 - \frac{V_b}{A_c} \right) \quad (2.16)$$

where L_1 is the distance along the stem of the hydrometer from the top of its bulb to the mark for a specific hydrometer reading, L_2 is the length of the hydrometer bulb (typically equal to 14 cm), V_b is the volume of the hydrometer bulb, and A_c is the cross-sectional area of the sedimentation cylinder (typically equal to 27.8 cm²).

Typically, L_1 is equal to 10.5 cm for a reading of $R = 0$, and 2.3 cm for a value of $R = 50$. Assuming a linear relationship between L_1 and R , it follows that for any value of R ,

$$L_1 = 10.5 - \frac{(10.5 - 2.3)}{50} R = 10.5 - 0.164R \quad (2.17)$$

where L_1 and R will have the same units, and the hydrometer reading R is corrected for the meniscus. In this case, the general expression for L becomes

$$L = 10.5 - 0.164R + \frac{1}{2} \left(14.0 - \frac{V_b}{A_c} \right) \quad (2.18)$$

where L and R will have units of cm, A_c will have units of cm², and V_b will have units of cm³.

2.3 PLASTICITY OF SOIL

As noted in [Section 2.1](#), when mixed with a limited amount of water, clays exhibit *plastic* (putty like) behavior characterized by deformation *without* complete rebound on removal of load. It is necessary to somehow quantify the plasticity of predominantly clayey soils; this will tell if a soil contains clay and if so, how much.

As also noted in [Section 2.1](#), particles classified as being “clay” on the basis of their size may not necessarily contain clay minerals. Nonclay soils typically contain quartz, feldspar, or mica particles that are small enough to be less than 0.002 mm (2 μm) in size. It is thus appropriate for particles smaller than 0.002 or 0.005 mm to be called “clay-size” particles rather than clay.

TABLE 2.4 Dimensions of Typical Clay Platelets^a

Clay Mineral	Ratio of Dimensions	Range of Lengths and Breadths ($\times 10^{-10}$ m)	Range of Thicknesses ($\times 10^{-10}$ m)
Kaolinite	10:10:1	1000–2000	100–1000
Illite	20:20:1	1000–5000	50–5000
Montmorillonite	100:100:1	1000–5000	10–50

^aHough, B.K., 1957. *Basic Soils Engineering*. The Ronald Press Company, New York, NY.

Unlike gravels and sands that consist exclusively of quartz and feldspar, clays are composed of silicate members that can have different mineralogy. The predominant clay minerals are *kaolinite*, *illite*, and *montmorillonite*.

Clay particles are plate-like (flakey); i.e., they have a sheet-like shape. Table 2.4 lists some typical dimensions for the above clay minerals.

The faces of clay particles are negatively charged, and their edges are positively charged (Figure 2.2A). As a result, clay particles attract water molecules (dipoles), creating layers of tightly bound *adsorbed water* (Figure 2.2B).

In light of the above discussion, it is evident that gradation alone (e.g., sieve analysis) is *insufficient* to classify fine-grained soils. This is because the behavior of clays is affected by (1) *Mineralogy* (it controls the particle surface characteristics), and (2) *Specific surface* (i.e., surface area per unit mass of a dry particle), which determines the particle shapes. Table 2.5 presents some approximate specific surface values.

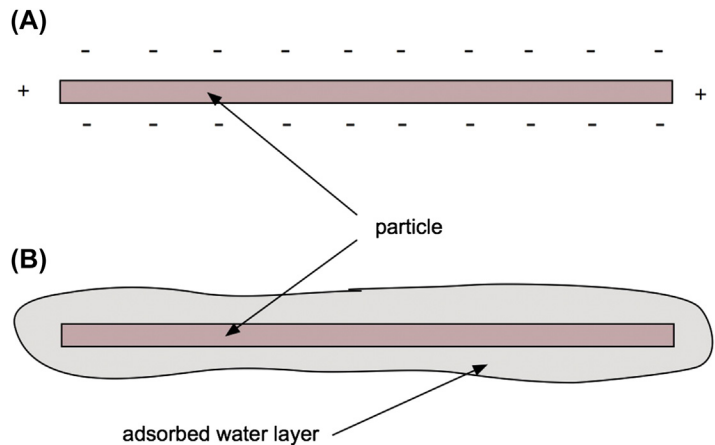


FIGURE 2.2 Schematic illustration of a typical clay particle: (A) surface charges, (B) adsorbed water layer.

TABLE 2.5 Approximate Values of Specific Surface for Common Clay Minerals^a

Soil Particle	Specific Surface (m ² /g)
Clean sand	0.0002
Kaolinite	10–20
Illite	65–100
Montmorillonite	Up to 840

^aGrim, R.E., 1959. Physico-chemical properties of soils: clay minerals. *Journal of the Soil Mechanics and Foundations Division, ASCE* 85 (SM2), 1–17.

Both mineralogy and specific surface are needed to properly classify clays. However, the measurement of both of these quantities is *not* practical for standard engineering applications. Instead, the nature of clays needs to be described by determining the degree of plasticity, which is attributed to the adsorbed water layers that surround a clay particle (Figure 2.2B).

2.4 ATTERBERG LIMITS

The Atterberg limits are a basic measure of the nature or consistency of a fine-grained soil. These limits correspond to specific values of moisture content (w). They were created by Albert Atterberg (1910) and later refined by Arthur Casagrande (1927). The Atterberg limits can be used to distinguish between a silt and a clay. In addition, they can distinguish between different types of silts and clays.

Depending on the moisture content of the soil, it may appear in four states: *solid*, *semisolid*, *plastic*, and *liquid* (Figure 2.3). In each state the consistency and behavior of a soil is different and thus so are its engineering properties. Thus, the boundary between each state can be defined based on a change in the soil's behavior.

2.4.1 Basic Definitions

The Atterberg limits consist of the following key values of moisture content:

- The *Liquid Limit* (LL) is the moisture content at which a fine-grained soil no longer flows like a liquid.
- The *Plastic Limit* (PL) is the moisture content at which a fine-grained soil can no longer be remolded without cracking.
- The *Shrinkage Limit* (SL) is the moisture content at which a fine-grained soil no longer changes volume upon drying—any loss of moisture is compensated by the entry of air into the pores.

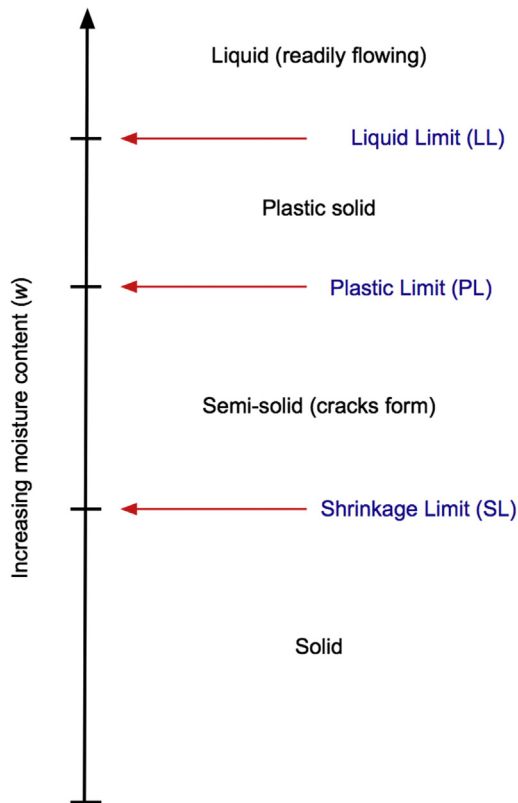


FIGURE 2.3 Schematic illustration of material states in a soil.

Remark: Whereas the LL and PL limits are *arbitrary* limits, the SL is a *definite* limit for a given soil.

Remark: The SL is useful for determining the swelling and shrinkage capacity of soils. In general, "soils that swell a lot will also shrink a lot."

Remark: Since the Atterberg limit tests are performed on remolded samples, the previous stress history of the soil is completely removed.

The Atterberg limits are useful in that they allow for soil behavior to be inferred. For example, soil having similar LL and plasticity index (PI) will

typically have similar strength/water content relationships. Thus, if such a relationship is known for one soil, it can be inferred for a soil with similar Atterberg limits.

2.4.2 Derived Limits

The values of these limits are computed from the Atterberg limits (i.e., from LL, PL, and SL). There is also a close relationship between the limits and properties of a soil such as *compressibility*, *permeability*, and *strength*. This is thought to be very useful because as the determination of Atterberg limits is relatively simple, it is more difficult to determine compressibility, permeability, and strength. The Atterberg limits are thus not only used to identify the soil's classification, but also allow for the use of empirical correlations for some other important engineering properties.

2.4.2.1 Plasticity Index

The PI is defined as the range of moisture contents over which the soil deforms plastically. The PI is thus defined to be the difference between the LL and the PL; i.e.,

$$PI = LL - PL \quad (2.19)$$

The PI thus is a measure of the plasticity of a soil. As such, the PI determines the amount and type of clay present in a soil. In general,

- Soils with a high PI tend to be clay,
- Those with a lower PI tend to be silt, and
- Those with a PI near zero tend to have little or no silt or clay (fines) present.

2.4.2.2 Liquidity Index

The liquidity index (LI) is used for scaling the natural water content of a soil sample to the limits. It is defined as follows:

$$LI = \frac{w - PL}{LL - PL} = \frac{w - PL}{PI} \quad (2.20)$$

where w is the natural water content. The LI serves as a measure of soil strength. In particular,

- If $LI < 0$, the soil is in a *semisolid* state characterized by high strength and brittle response characterized by sudden fracture of the soil.
- If $0 < LI < 1$, the soil is in a *plastic* state characterized by intermediate strength; it deforms like a plastic material.
- If $LI > 1$, the soil is in a *liquid* state characterized by low strength; it deforms like a viscous fluid.

Sensitive clays are ones that, when remolded, can be transformed into a viscous form that will flow like a fluid. For such soils the in situ moisture content may be greater than the LL, implying that $LI > 1$.

2.4.2.3 Activity

If a small amount of sand is added to a clay, the LL and PL for the soil will decrease. The LL and PL are thus functions of not only the type of clay present, but also the amount of clay present.

Skempton³ observed that the PI of a given soil increased approximately linearly with the percent of clay-size fraction. He defined the *activity* of a clay as

$$A = \frac{PI}{\text{percent of clay} - \text{size fraction}} \quad (2.21)$$

where the percent of clay-size fraction equals the percent by weight finer than 0.002 mm.

From the activity it is often possible to predict the dominant clay mineral present in a soil sample, and thus *indirectly* the specific surface. Table 2.6 gives ranges of activity values associated with the three most common clay minerals.

- For $0.75 \leq A \leq 1.25$, the clay is considered to be “normal”.
- For $A < 0.75$, the clay is considered to be “inactive”.
- For $A > 1.25$, the clay is considered to be “active”.

Remark: Soils with high activity are very reactive chemically.

Remark: High activity signifies large volume increase in a soil that is wetted and large shrinkage when it is dried.

Remark: Graphically, the activity is the slope of a straight line drawn on a figure with PI as the ordinate and percentage of clay-size fraction as the abscissa. Such lines will not always pass through the origin.

3. Skempton, A.W., 1953. The Colloidal Activity of Clays. In: Proceedings of the Third International Conference on Soil Mechanics and Foundation Engineering (I), 57–61.

TABLE 2.6 Typical Values of Liquid Limit, Plastic Limit, and Activity

Clay Mineral	LL	PL	Activity (A)
Kaolinite	35–100	20–40	0.3–0.5
Illite	60–12	35–60	0.5–1.3
Montmorillonite	100–900	50–100	1.5–7.0

2.5 SOIL CLASSIFICATION

To describe soils more effectively based on particle size distributions, plasticity, etc., several soil classifications schemes have been developed. In this section the Unified Soil Classification System (USCS) is briefly reviewed. Additional details pertaining to this soil classification system are found in most soil mechanics textbooks.

The USCS uses the following prefix symbols to describe the various soil groups:

- G = gravel or gravelly soil.
- S = sand or sandy soil.
- M = inorganic silt.
- C = inorganic clay.
- O = organic⁴ silt or clay.
- Pt = peat, muck, and other highly organic soils.

The above symbols are combined with the following descriptor symbols:

- W = well graded.
- P = poorly graded.
- L = low plasticity ($LL < 50\%$).
- H = high plasticity ($LL \geq 50\%$).

Thus, SP denotes a poorly graded sand; CL denotes a clayey soil with low plasticity, etc. Such combinations of symbols constitute the following group symbols:

- For *coarse-grained* soils (i.e., those with more than 50% retained on the No. 200 sieve):

The following symbols are used to classify *gravels* (“G” prefix):
 GW = well-graded gravel.

4. Organic matter is matter that comes from a once-living organism; it is capable of decay, or is the product of decay, or is composed of organic compounds. In a decaying material W_s and V_s may decrease with time.

GP = poorly graded gravel.

GM = silty gravel (i.e., nonclay fines).

GC = clayey gravel.

In addition, the description of gravels with 5% to 12% fines requires the following dual symbols:

GW-GM = well-graded gravel with silt.

GW-GC = well-graded gravel with clay.

GP-GM = poorly graded sand with silt.

GP-GC = poorly graded sand with clay.

The following symbols are used to classify *sand* (“S” prefix):

SW = well-graded sand.

SP = poorly graded sand.

SM = silty sand (i.e., nonclay fines).

SC = clayey sand.

In addition, the description of sand with 5% to 12% fines requires the following dual symbols:

SW-SM = well-graded sand with silt.

SW-SC = well-graded sand with clay.

SP-SM = poorly graded sand with silt.

SP-SC = poorly graded sand with clay.

- For *fine-grained* soils (i.e., those with 50% or more passing No. 200 sieve):

For silts and clays with $LL < 50\%$:

ML = inorganic silt.

CL = inorganic clay.

OL = organic silt and organic clay.

For silts and clays with $LL \geq 50\%$:

MH = inorganic silt.

CH = inorganic clay.

OH = organic silt and organic clay.

In addition, fines may be classified as CL-ML.

Fine-grained soils (i.e., silts and clays) are classified according to their Atterberg limits and whether or not they contain organic matter. Figure 2.4 shows the plasticity chart developed by Casagrande⁵, based on the results of tests performed on soils from throughout the world.

In Casagrande’s plasticity chart a value of $LL = 50\%$ delineates low plasticity silts and clays from high plasticity ones. In Figure 2.4 this dividing value is represented by the so-called “B-line”.

5. Casagrande, A., 1948. Classification and Identification of Soils. Transactions, ASCE 113, 901–930.

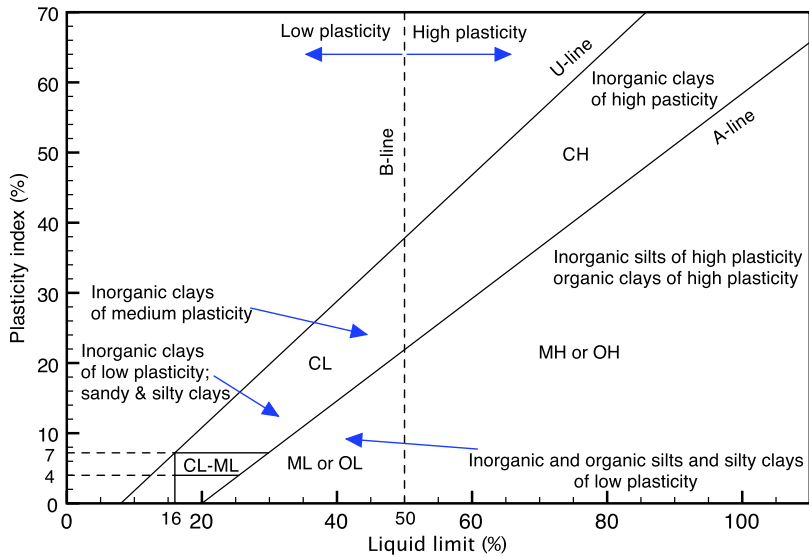


FIGURE 2.4 Classification of fine-grained soils on Casagrande's plasticity chart.

The "A-line" in Figure 2.4 is defined by

$$PI = 0.73(LL - 20) \quad (2.22)$$

where LL and PI are expressed in percent. It represents the boundary between inorganic and organic soils.

The "U-line" in Figure 2.4 is defined by

$$PI = 0.9(LL - 8) \quad (2.23)$$

where LL and PI are again expressed in percent. It represents the upper limit of the correlation between PI and LL for any currently known fine-grained soil.

EXAMPLE PROBLEM 2.1

General Remarks

This example problem illustrates the manner in which data from sieve analyses are manipulated so as to construct a particle-size distribution curve.

Problem Statement

A sieve analysis was performed on a clean gravelly sand. The first three columns of Table Ex. 2.1A summarize the data from this analysis. Given these data,

TABLE EX. 2.1A Results of Sieve Analyses Performed on a Clean Gravelly Sand

Sieve Number	Sieve Opening (mm)	Mass of Soil Retained (g)	Cumulative Mass of Soil Retained on Sieve (g)	Percent Finer Than a Given Sieve Size
4	4.75	0	0	100.0
7	2.80	492	0 + 492 = 492	77.7
18	1.00	898	492 + 898 = 1390	37.0
40	0.425	295	1390 + 295 = 1685	23.7
50	0.355	213	1685 + 213 = 1898	14.0
80	0.180	130	1898 + 130 = 2028	8.2
170	0.090	160	2028 + 160 = 2188	0.9
Pan	—	20	2188 + 20 = 2208	0.0
Total	—	2208	—	—

- a) Plot the particle-size distribution curve, and,
- b) Determine the coefficient of uniformity (C_u) and the coefficient of gradation (C_c).

Solution

- a) The fourth column shows how the cumulative mass of soil retained on a given sieve is computed. In the fifth column is the percent finer than a given sieve size, which is computed as follows:

$$\%_{\text{finer}} = \left(\frac{M_{\text{total}} - M_{\text{cumulative}}}{M_{\text{total}}} \right) * 100\% \quad (2.1.1)$$

where $M_{\text{cumulative}}$ and M_{total} are the cumulative mass of soil retained on sieve and the total mass, respectively.

Remark: The above calculations are particularly well suited for being performed using a spreadsheet.

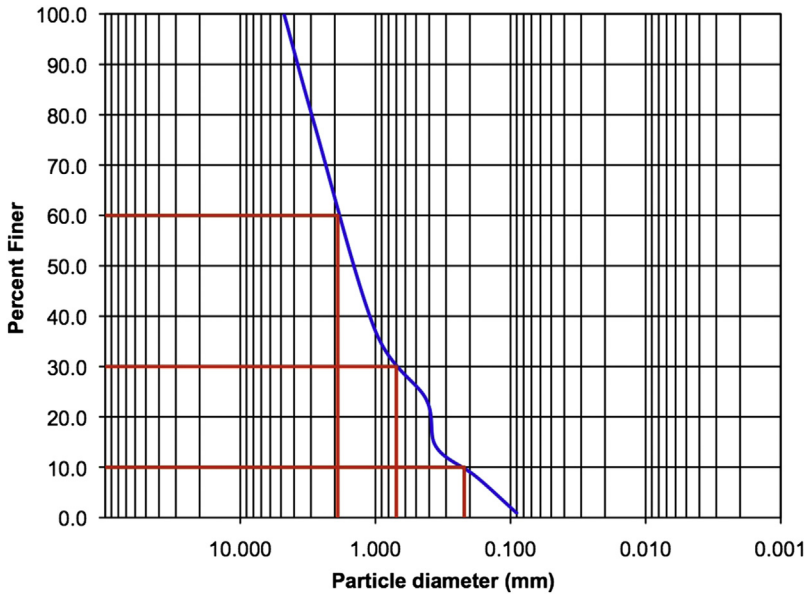


FIGURE EX. 2.1 Particle size distribution for a clean gravelly sand.

Figure Ex. 2.1 shows the particle size distribution for the clean gravelly sand.

From Figure Ex. 2.1 the diameters corresponding to 60% finer (D_{60}), 30% finer (D_{30}), and 10% finer (D_{10}) are found to be approximately 1.90, 0.70, and 0.22 mm, respectively. The coefficient of uniformity is thus

$$C_u = \frac{D_{60}}{D_{10}} = \frac{1.90}{0.22} = \mathbf{8.6} \quad (2.1.2)$$

Finally, coefficient of gradation is

$$C_c = \frac{(D_{30})^2}{D_{10}D_{60}} = \frac{(0.70)^2}{(0.22)(1.90)} = \mathbf{1.17} \quad (2.1.3)$$

EXAMPLE PROBLEM 2.2

General Remarks

This example problem further illustrates the manner in which particle-size distribution curves are drawn, and the coefficients that are computed from them.

TABLE EX. 2.2 Results of Sieve Analyses Performed on Two Soils

Glacial Till		Inorganic Silt	
Particle Size (mm)	Percent Finer	Particle Size (mm)	Percent Finer
0.295	97	0.074	96
0.147	94	0.050	89
0.074	69	0.030	72
0.055	48	0.015	47
0.035	22	0.095	34
0.025	6	0.0045	18
0.015	1	0.0015	8

Problem Statement

Sieve analyses were performed on two soils. [Table Ex. 2.2](#) summarizes the results of these analyses. For each soil,

- Plot the particle-size distribution curve, and,
- Determine the coefficient of uniformity (C_u) and the coefficient of gradation (C_c).

Solution

[Figure Ex. 2.2A](#) shows the particle size distribution for the glacial till.

From [Figure Ex. 2.2A](#) the diameters corresponding to 60% finer (D_{60}), 30% finer (D_{30}), and 10% finer (D_{10}) are found to be 0.300, 0.040, and 0.007 mm, respectively. The coefficient of uniformity is thus

$$C_u = \frac{D_{60}}{D_{10}} = \frac{0.300}{0.007} = \mathbf{42.9} \quad (2.2.1)$$

Finally, coefficient of gradation is

$$C_c = \frac{(D_{30})^2}{D_{10}D_{60}} = \frac{(0.040)^2}{(0.007)(0.300)} = \mathbf{0.76} \quad (2.2.2)$$

[Figure Ex. 2.2B](#) shows the particle size distribution for the inorganic silt.

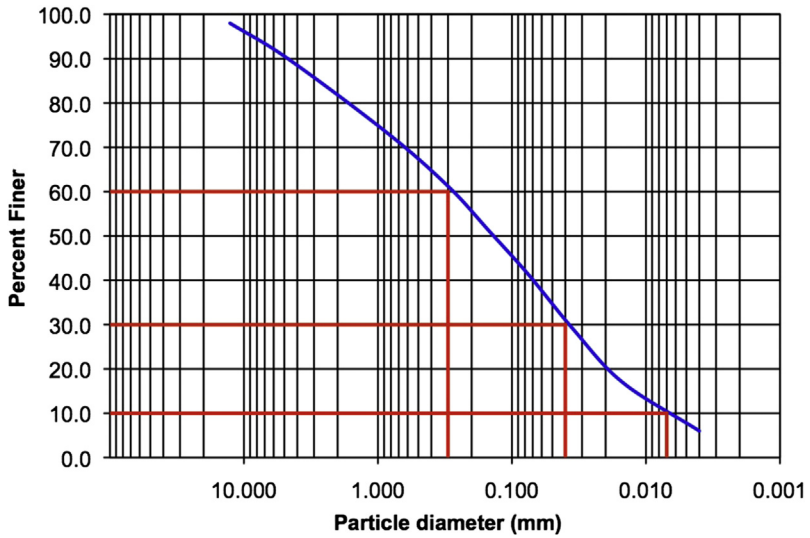


FIGURE EX. 2.2A Particle size distribution for a glacial till.

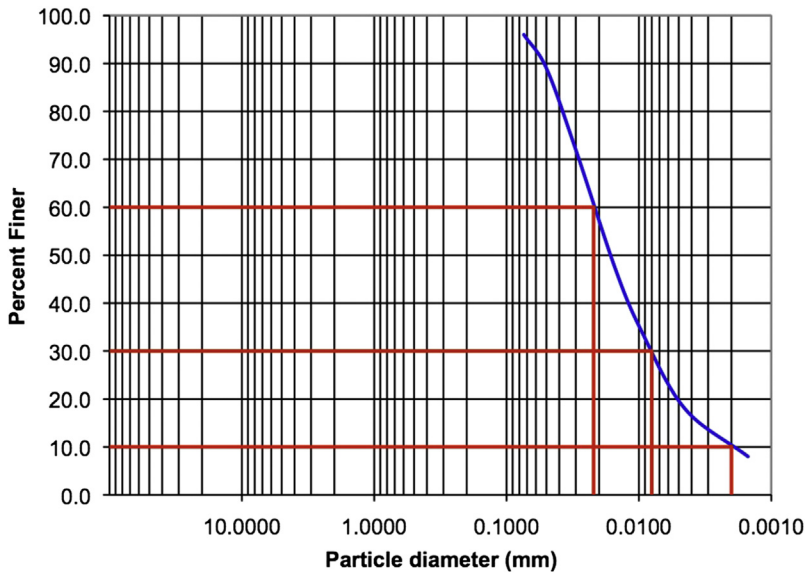


FIGURE EX. 2.2B Particle size distribution for an inorganic silt.

From Figure Ex. 2.2B the diameters corresponding to 60% finer (D_{60}), 30% finer (D_{30}), and 10% finer (D_{10}) are found to be 0.022, 0.008, and 0.002 mm, respectively. The coefficient of uniformity is thus

$$C_u = \frac{D_{60}}{D_{10}} = \frac{0.022}{0.002} = \mathbf{11.0} \quad (2.2.3)$$

Finally, coefficient of gradation is

$$C_c = \frac{(D_{30})^2}{D_{10}D_{60}} = \frac{(0.008)^2}{(0.002)(0.022)} = \mathbf{1.46} \quad (2.2.4)$$

EXAMPLE PROBLEM 2.3

General Remarks

This example problem investigates the relationship between the initial suspension density (ϕ_0), the specific gravity of solids (G_s) and the initial concentration in a hydrometer analysis.

Problem Statement

Since the total weight (W) and total volume (V) of a sample are known, and since all of the soil is in suspension at the outset of a hydrometer test, the initial concentration is likewise known. Determine (a) a general relationship between ϕ_0 , W , and V , and (b) the initial suspension density for a liter suspension containing 49 g of soil with a specific gravity of solids equal to 2.73.

Solution

- a) Since the sample is saturated, its volume and weight are related in the manner shown in Figure Ex. 2.3.

From the definition of the specific gravity of solids,

$$G_s = \frac{W_s}{V_s \gamma_w} \Rightarrow V_s = \frac{W_s}{G_s \gamma_w} \quad (2.3.1)$$

Thus,

$$V_w = V - V_s = V - \frac{W_s}{G_s \gamma_w} \quad (2.3.2)$$

From the definition of the unit weight of water,

$$\gamma_w = \frac{W_w}{V_w} \Rightarrow W_w = \gamma_w V_w = \gamma_w \left(V - \frac{W_s}{G_s \gamma_w} \right) = V \gamma_w - \frac{W_s}{G_s} \quad (2.3.3)$$

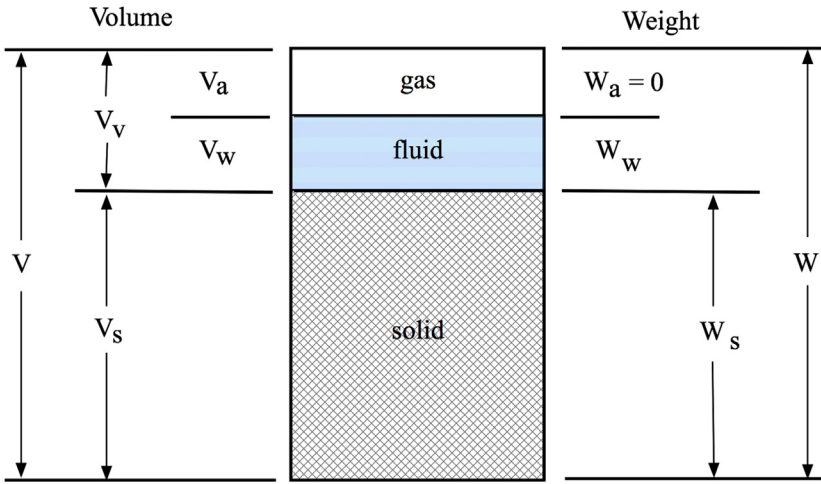


FIGURE EX. 2.3 Phase relationship associated with a hydrometer test.

The initial suspension density is thus

$$\phi_0 = \frac{W_s + W_w}{V} = \frac{W_s + \left(V\gamma_w - \frac{W_s}{G_s} \right)}{V} = \frac{W_s}{V} \left(1 - \frac{1}{G_s} \right) + \gamma_w \quad (2.3.4)$$

Recalling the definition of the dry unit weight given in Eq. (1.9); i.e.,

$$\gamma_d = \frac{W_s}{V} \quad (2.3.5)$$

The initial suspension density can likewise be written as

$$\phi_0 = \gamma_d \left(1 - \frac{1}{G_s} \right) + \gamma_w \quad (2.3.6)$$

- b) The initial suspension density for a liter suspension containing 49 g of soil with a specific gravity of solids equal to 2.73 is thus

$$\phi_0 = \frac{(49 \text{ g})}{(1000 \text{ cm}^3)} \left(1 - \frac{1}{2.73} \right) + (1.0 \text{ g/cm}^3) = \mathbf{1.031 \text{ g/cm}^3} \quad (2.3.7)$$

EXAMPLE PROBLEM 2.4

General Remarks

This example problem illustrates the sample calculations that are performed for a hydrometer analysis that was run using an ASTM 152-H hydrometer (Figure Ex. 2.4A).

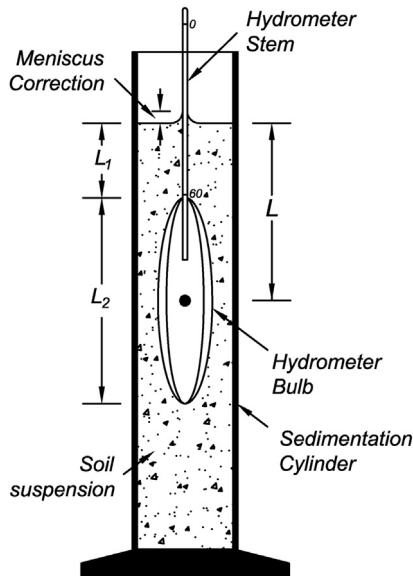


FIGURE EX. 2.4A Hydrometer in a soil suspension.

Problem Statement

Given the soil data shown in Table Ex. 2.4A, the hydrometer calibration information in Table Ex. 2.4B, and the results of a hydrometer test shown in Table Ex. 2.4C, draw the grain size distribution curve for the soil.

TABLE EX. 2.4A Soil Data Associated With a Hydrometer Analysis

Mass Retained on No. 200 Sieve	0.0 g
Mass in suspension (M_s)	48.0 g
Specific gravity (G_s)	2.70
Meniscus correction (F_m)	0.5 g/L
Dispersing agent correction (F_z)	4.0 g/L
Diameter of sedimentation cylinder	5.95 cm
Hydrometer bulb volume	67 cm ³

TABLE EX. 2.4B Hydrometer Test Data

Time (min)	Hydrometer Reading (g/L)	Temperature (°C)
1	38	22.0
2	33	22.0
3	30	21.5
4	28	21.5
8	25	21.5
15	23	21.0
30	21	21.0
60	19	21.0
240	15	19.5
900	12	18.5

Solution

Some preliminary calculations are first performed. The specific gravity correction is

$$a = \frac{1.65G_s}{2.65(G_s - 1)} = \frac{1.65(2.70)}{2.65(2.70 - 1)} = 0.989 \quad (2.4.1)$$

The cross-sectional area of the sedimentation cylinder is

$$A_c = \frac{\pi}{4}(5.95 \text{ cm})^2 = 27.8 \text{ cm}^2 \quad (2.4.2)$$

The general expression for the effective length is now specialized for the ASTM 152-H hydrometer, as follows:

$$L_1 = 10.5 - \left(\frac{10.5 - 2.3}{50} \right) R_{CL}; \quad L_2 = 14 \text{ cm} \quad (2.4.3)$$

Then,

$$\begin{aligned} L &= L_1 + \frac{1}{2} \left(L_2 - \frac{V_b}{A_c} \right) = 10.5 - 0.164R_{CL} + \frac{1}{2} \left(14.0 - \frac{67 \text{ cm}^3}{27.8 \text{ cm}^2} \right) \\ &= 16.295 - 0.164R_{CL} \end{aligned} \quad (2.4.4)$$

TABLE EX. 2.4C Results of Hydrometer Analysis Calculations

Time (min)	R (g/L)	Temperature ($^{\circ}\text{C}$)	F_t	R_{CP} (g/L)	% Finer	R_{CL} (g/L)	L (cm)	K	D (mm)
1	38	22.0	0.485	34.49	71.05	38.50	9.98	0.01312	0.04145
2	33	22.0	0.485	29.49	60.75	33.50	10.80	0.01312	0.03049
3	30	21.5	0.364	26.36	54.31	30.50	11.29	0.01312	0.02545
4	28	21.5	0.364	24.36	50.19	28.50	11.62	0.01312	0.02236
8	25	21.5	0.364	21.36	44.01	25.50	12.11	0.01312	0.01614
15	23	21.0	0.243	19.24	39.64	23.50	12.44	0.01312	0.01195
30	21	21.0	0.243	17.24	35.52	21.50	12.77	0.01312	0.00856
60	19	21.0	0.243	15.24	31.40	19.50	13.10	0.01312	0.00613
240	15	19.5	-0.121	10.88	22.41	15.50	13.75	0.01312	0.00314
900	12	18.5	-0.364	7.64	15.73	12.50	14.24	0.01312	0.00165

Notes: Meniscus correction (F_m) = 0.5 g/L, Dispersing agent correction (F_z) = 4.0 g/L.

For the first hydrometer reading, $R = 38$ g/L at a temperature of 22.0°C . The temperature correction for this reading is thus

$$F_T = -4.85 + 0.2425T = -4.85 + 0.2425(22.0) = 0.485 \quad (2.4.5)$$

The corrected hydrometer reading for percent finer is then

$$R_{CP} = R + F_T - F_z = 38 + 0.485 - 4.0 = 34.49\text{g/L} \quad (2.4.6)$$

The percent finer is next computed as follows:

$$\%finer = \left(\frac{aR_{CP}}{M_s} \right) * 100\% = \left(\frac{(0.989)(34.49\text{g/L})}{48.0\text{g}} \right) * 100\% = \mathbf{71.05\%} \quad (2.4.7)$$

The corrected hydrometer reading for determination of effective length is next computed as follows:

$$R_{CL} = R + F_m = 38 + 0.5 = 38.5 \text{ g/L} \quad (2.4.8)$$

The effective length corresponding to R_{CL} is then

$$L = 16.295 - 0.164(38.5\text{g/L}) = 9.981 \text{ cm} \quad (2.4.9)$$

The value of K must next be computed. At 22.0°C the viscosity of water is equal to 9.754×10^{-6} (g s)/cm², thus

$$K = \sqrt{\frac{30\eta}{(G_s - 1)}} = \sqrt{\frac{30(9.754 \times 10^{-6}\text{g s/cm}^2)}{(2.70 - 1)}} = 1.312 \times 10^{-2} \quad (2.4.10)$$

Finally, the particle diameter is computed as follows:

$$D = K\sqrt{\frac{L}{t}} = (1.312 \times 10^{-2})\sqrt{\frac{9.981 \text{ cm}}{1.0 \text{ min}}} = \mathbf{0.0414 \text{ mm}} \quad (2.4.11)$$

The percent finer value computed above, and the particle diameter, defines a point on the grain size distribution curve. [Table Ex. 2.4C](#) summarizes the results computed for all of the hydrometer readings.

[Figure Ex. 2.4B](#) shows the particle size distribution curve that was developed using data from the above hydrometer analysis.

EXAMPLE PROBLEM 2.5

General Remarks

This example problem illustrates how the liquid limit is computed using data obtained from a Casagrande device.

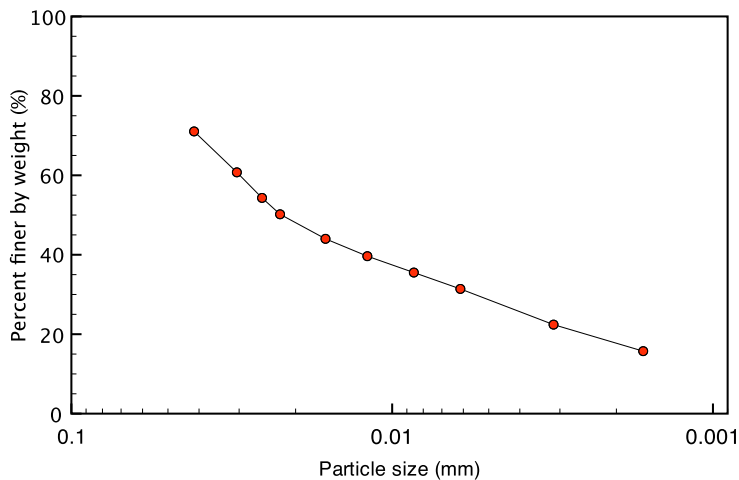


FIGURE EX. 2.4B Particle size distribution curve from hydrometer analysis.

Problem Statement

A series of Atterberg limit tests was performed on samples of a gray–brown silty clay. Table Ex. 2.5A summarizes the results of four liquid limit tests. Using this data,

- a) Draw the *flow curve* for the soil.
- b) Determine the *flow index* for the soil.
- c) Calculate the *liquid limit*.
- d) Calculate the *plasticity index* for the soil.

In the same series of Atterberg limit tests, the average plastic limit (w_{PL}) was found to be 38.5%.

TABLE EX. 2.5A Liquid Limit Data Obtained From a Casagrande Device				
Trial Number	1	2	3	4
Number of blows	24	37	27	22
Weight of wet sample + container (g)	22.50	25.94	25.86	25.27
Weight of dry sample + container (g)	18.89	21.10	21.59	20.96
Weight of container (g)	14.12	14.61	15.82	15.34

Solution

Consider trial number 1. The weight of the pore fluid is first computed as follows:

$$W_w = 22.50 - 18.89 = 3.61 \text{ g} \quad (2.5.1)$$

The weight of the solid phase is next computed

$$W_s = 18.89 - 14.12 = 4.77 \text{ g} \quad (2.5.2)$$

The moisture content for trial number 1 is thus

$$w = \left(\frac{3.61 \text{ g}}{4.77 \text{ g}} \right) * 100\% = 75.7\% \quad (2.5.3)$$

Similar calculations are performed for the remaining three trials. [Table Ex 2.5B](#) lists the complete results for all four trials.

- a) To construct the flow curve, the moisture content is plotted versus the number of blows (in logarithmic scale). [Figure Ex. 2.5](#) shows the resulting curve.
- b) The *flow index* for the soil is equal to the absolute value of the slope of the flow curve. As such, it is equal to **0.146**.
- c) The liquid limit is then the moisture content corresponding to 25 blows. In this case,

$$w_{LL} \approx \mathbf{75.5\%} \quad (2.5.4)$$

- d) The *plasticity index* (I_p) is the difference between the liquid limit and the plastic limit. It represents the range of moisture content over which the

TABLE EX. 2.5B Computed Values Obtained Using Liquid Limit Data Obtained From a Casagrande Device

Trial Number	1	2	3	4
Number of blows	24	37	32	22
Weight of wet sample + container (g)	22.50	25.94	25.86	25.27
Weight of dry sample + container (g)	18.89	21.10	21.59	20.96
Weight of container (g)	14.12	14.61	15.82	15.34
Weight of pore fluid (g)	3.61	4.84	4.27	4.31
Weight of solid phase (g)	4.77	6.49	5.77	5.62
Moisture content (%)	75.7	74.6	74.0	76.7

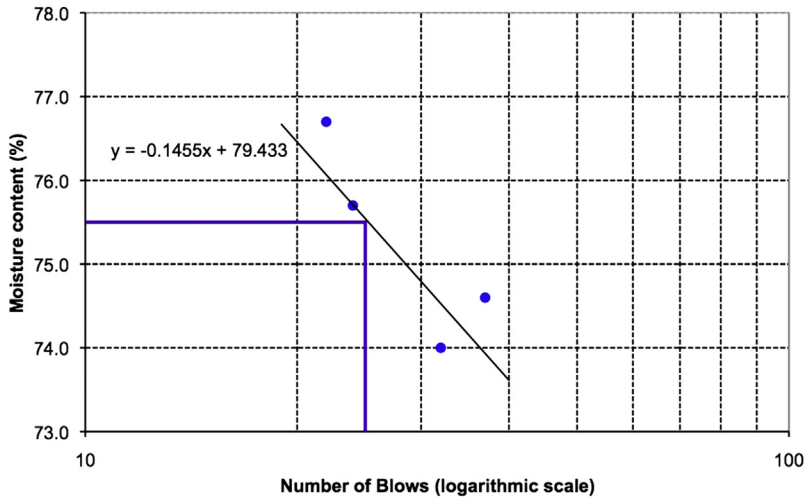


FIGURE EX. 2.5 Flow curve obtained from a Casagrande device.

soil behaves plastically (i.e., it is moldable). The value of I_p is important in classifying fine-grained soils. For the soil in question, the plasticity index is

$$I_p = w_{LL} - w_{PL} = 75.5 - 38.5 = \mathbf{37.0\%} \quad (2.5.5)$$

EXAMPLE PROBLEM 2.6

General Remarks

The liquid limit can also be computed using data obtained from a cone penetrometer.

Problem Statement

Given data obtained as a result of a liquid limit test (Table Ex. 2.6),

- Draw the *flow curve* for the soil.
- Determine the *flow index* for the soil.
- Calculate the *liquid limit*.
- Calculate the *plasticity index* for the soil.

In the same series of Atterberg limit tests, the plastic limit (w_{PL}) was found to be 27.2%.

TABLE EX. 2.6 Liquid Limit Data Obtained From a Cone Penetrometer

Specimen	Moisture Content (%)	Penetration (mm)
1	37.2	16
2	39.1	27
3	44.3	36
4	46.0	54
5	49.8	69

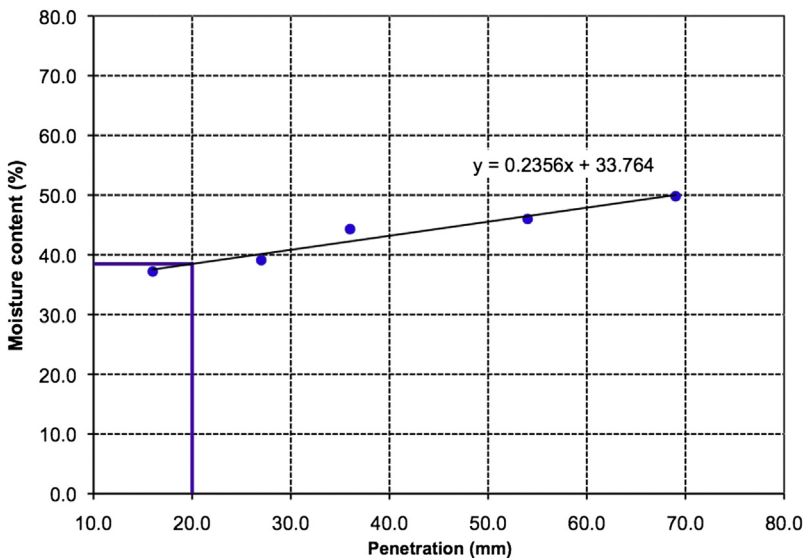
Solution

- a) To construct the flow curve, the moisture content is plotted versus the penetration. Figure Ex. 2.6 shows the resulting curve.
- b) The *flow index* for the soil is equal to the absolute value of the slope of the flow curve. As such, it is equal to **0.236**.
- c) The liquid limit is then the moisture content corresponding to 20 mm penetration. Thus,

$$w_{LL} \approx \mathbf{38.5\%} \quad (2.6.1)$$

- d) Finally, the plasticity index is

$$I_p = w_{LL} - w_{PL} = 38.5 - 27.2 = \mathbf{11.3\%} \quad (2.6.2)$$

**FIGURE EX. 2.6** Flow curve obtained from a cone penetrometer.

EXAMPLE PROBLEM 2.7

General Remarks

The shrinkage limit is investigated in this problem. As a saturated soil is slowly dried, capillary menisci form between the individual soil particles. As a result, the interparticle (effective) stresses increase, and the soil decreases in volume. A point is eventually reached where this volume change stops, even though the degree of saturation is still essentially 100%. The moisture content at which this occurs is defined as the *shrinkage limit* (w_{SL}), which, as noted above, is one of the Atterberg limits.

Figure Ex 2.7 shows the relation between the total weight and total volume of the soil. In this figure W_i and V_i denote the initial weight and volume of the saturated soil. The weight and volume associated with the shrinkage limit are W_{SL} and V_{dry} , respectively. Finally, W_s and V_s again denote the weight and volume of the solid phase. The values of W_i , V_i , W_s , and V_{dry} are easily measured in the laboratory.

By definition, the shrinkage limit is given by

$$w_{SL} = \left(\frac{W_{SL} - W_s}{W_s} \right) * 100\% \quad (2.7.1)$$

Since the soil is essentially saturated, it follows that

$$W_i - W_{SL} = \gamma_w (V_i - V_{dry}) \Rightarrow W_{SL} = W_i - \gamma_w (V_i - V_{dry}) \quad (2.7.2)$$

Substituting this result into the previous equation gives the following expression for w_{SL} :

$$\begin{aligned} w_{SL} &= \left[\frac{W_i - \gamma_w (V_i - V_{dry}) - W_s}{W_s} \right] * 100\% \\ &= \left[\frac{W_i - W_s - \gamma_w (V_i - V_{dry})}{W_s} \right] * 100\% = w_i - \left[\frac{\gamma_w (V_i - V_{dry})}{W_s} \right] * 100\% \end{aligned} \quad (2.7.3)$$

where w_i is expressed in percent.

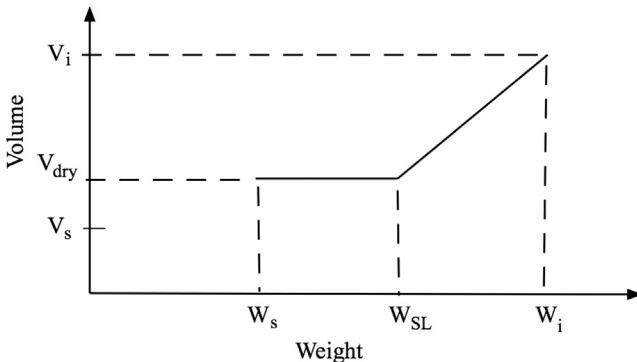


FIGURE EX. 2.7 Relationship between volume and weight associated with the shrinkage limit.

Problem Statement

To illustrate the use of the above equation, consider the following laboratory data:

- Weight of wet sample + container = 40.64 g
- Weight of dry sample + container = 30.42 g
- Weight of container = 18.36 g
- Initial volume of soil (V_i) = 15.07 cm³
- Volume of soil after drying (V_{dry}) = 7.12 cm³

Compute the shrinkage limit for the above data.

Solution

The weight of the pore fluid is thus

$$W_w = 40.64 - 30.42 = 10.22 \text{ g} \quad (2.7.4)$$

The weight of the solid is

$$W_s = 30.42 - 18.36 = 12.06 \text{ g} \quad (2.7.5)$$

The initial moisture content is thus (recall Eq. 1.14)

$$w_i = \left(\frac{W_w}{W_s} \right) * 100\% = \left(\frac{10.22 \text{ g}}{12.06 \text{ g}} \right) * 100\% = 84.7\% \quad (2.7.6)$$

Using the above values in Eq. (2.7.3), the shrinkage limit is thus

$$\begin{aligned} w_{\text{SL}} &= w_i - \left[\frac{\gamma_w (V_i - V_{\text{dry}})}{W_s} \right] * 100\% \\ &= 84.7\% - \left[\frac{(1.0 \text{ g/cm}^3)(15.07 - 7.12 \text{ cm}^3)}{12.06 \text{ g}} \right] * 100\% \\ &= \mathbf{18.8\%} \end{aligned} \quad (2.7.7)$$

Two quantities related to the shrinkage limit are sometimes also computed. The first quantity is the linear shrinkage ratio

$$\text{LS} = 1 - \left(\frac{V_{\text{dry}}}{V_i} \right)^{\frac{1}{3}} \quad (2.7.8)$$

The second quantity is the shrinkage ratio,

$$\text{SR} = \frac{W_s}{V_{\text{dry}} \gamma_w} \quad (2.7.9)$$

For the given data, Eqs. (2.7.8) and (2.7.9) give

$$\text{LS} = 1 - \left(\frac{7.12 \text{ cm}^3}{15.07 \text{ cm}^3} \right)^{\frac{1}{3}} = \mathbf{0.22} \quad (2.7.10)$$

$$SR = \frac{W_s}{V_{\text{dry}} \gamma_w} = \frac{12.06 \text{ g}}{(7.12 \text{ cm}^3)(1.0 \text{ g/cm}^3)} = \mathbf{1.69} \quad (2.7.11)$$

The next several problems give additional insight into some computations related to the Atterberg limits.

EXAMPLE PROBLEM 2.8

General Remarks

This example problem illustrates the computation of the shrinkage limit.

Problem Statement

In a test to determine the shrinkage limit of a cohesive soil, after drying, a sample had a volume of 50 cm^3 and a weight of 88.0 g . The specific gravity of solids was 2.71 . Determine the shrinkage limit (w_{SL}).

Solution

Referring to [Figure Ex 2.7](#), from the given information $W_s = 88.0 \text{ g}$. From the definition of the specific gravity of solids,

$$G_s = \frac{W_s}{V_s \gamma_w} \Rightarrow V_s = \frac{W_s}{G_s \gamma_w} = \frac{88.0 \text{ g}}{(2.71)(1.0 \text{ g/cm}^3)} = 32.47 \text{ cm}^3 \quad (2.8.1)$$

Since the soil is assumed to be saturated, the volume of the pore fluid is thus

$$V_w = 50.0 \text{ cm}^3 - V_s = 50.0 - 32.47 = 17.53 \text{ cm}^3 \quad (2.8.2)$$

The weight of the pore fluid at the shrinkage limit is thus

$$W_w = V_w \gamma_w = (17.53 \text{ cm}^3)(1.0 \text{ g/cm}^3) = 17.53 \text{ g} \quad (2.8.3)$$

The shrinkage limit is thus

$$w_{\text{SL}} = \left(\frac{W_w}{W_s} \right) * 100\% = \left(\frac{17.53 \text{ g}}{88.0 \text{ g}} \right) * 100\% = \mathbf{21.9\%} \quad (2.8.4)$$

EXAMPLE PROBLEM 2.9

General Remarks

This example problem illustrates some of the calculations that are performed using Atterberg limits.

Problem Statement

The liquid limit (w_{LL}) of a clay soil is 54% , and its plasticity index (I_p) is 15% .

(a) What is the plastic limit (w_{PL}) of the soil? (b) In what state of consistency

is this material at a moisture content of 40%? (c) At the minimum volume reached during shrinkage, a sample of this soil has a void ratio of 0.87. If the specific gravity of solids is equal to 2.72, compute the shrinkage limit (w_{SL}).

Solution

- a) The plastic limit is computed from the definition of the plasticity index as follows:

$$I_P = w_{LL} - w_{PL} \Rightarrow w_{PL} = w_{LL} - I_P = 54 - 15 = \mathbf{39\%} \quad (2.9.1)$$

- b) Since $40\% > 39\%$, it follows that the soil is still **plastic**.

- c) Since, at the minimum volume reached during shrinkage the soil is assumed to be saturated, the moisture content associated with the shrinkage limit is thus

$$w_{SL} = \left(\frac{e}{G_s} \right) * 100\% = \left(\frac{0.87}{2.72} \right) * 100\% = \mathbf{32.0\%} \quad (2.9.2)$$

EXAMPLE PROBLEM 2.10

General Remarks

This example problem illustrates some of the calculations involving the shrinkage limit.

Problem Statement

A clay soil has a liquid limit of 60% and a shrinkage limit of 25%. If a specimen of this soil shrinks from an initial volume of 10.0 cm^3 at the liquid limit to a volume of 6.39 cm^3 at the shrinkage limit, what is the specific gravity of solids (G_s)?

Letting w_{LL} and w_{SL} denote the liquid and shrinkage limits, respectively, the moisture content for both limits is thus

$$w_{LL} = \frac{W_{wL}}{W_s} \Rightarrow W_{wL} = w_{LL} W_s \quad (2.10.1)$$

$$w_{SL} = \frac{W_{wS}}{W_s} \Rightarrow W_{wS} = w_{SL} W_s \quad (2.10.2)$$

Since the soil is assumed to be saturated, the change in the weight of the fluid phase is given by

$$\Delta W_w = (W_{wL} - W_{wS}) = \gamma_w (V_{LL} - V_{dry}) \quad (2.10.3)$$

Substituting for the weight of the fluid gives

$$\Delta W_w = (w_{LL} - w_{SL}) W_s = \gamma_w (V_{LL} - V_{dry}) \Rightarrow W_s = \frac{\gamma_w (V_{LL} - V_{dry})}{w_{LL} - w_{SL}} \quad (2.10.4)$$

The next quantity required to compute G_s is the volume of the solid phase (V_s). The volume of the pore fluid at the shrinkage limit is first computed

$$V_{w_s} = \frac{W_{w_s}}{\gamma_w} = \frac{w_{SL} W_s}{\gamma_w} = \frac{w_{SL}(V_{LL} - V_{dry})}{w_{LL} - w_{SL}} \quad (2.10.5)$$

Since the soil is saturated,

$$V_s = V_{dry} - V_{w_s} = V_{dry} - \frac{w_{SL}(V_{LL} - V_{dry})}{w_{LL} - w_{SL}} = \frac{V_{dry}w_{LL} - V_{LL}w_{SL}}{w_{LL} - w_{SL}} \quad (2.10.6)$$

The specific gravity of solids is thus

$$G_s = \frac{W_s}{V_s \gamma_w} = \frac{\gamma_w(V_{LL} - V_{dry})}{(w_{LL} - w_{SL})} \frac{(w_{LL} - w_{SL})}{(V_{dry}w_{LL} - V_{LL}w_{SL})\gamma_w} = \frac{V_{LL} - V_{dry}}{V_{dry}w_{LL} - V_{LL}w_{SL}} \quad (2.10.7)$$

Substituting for all of the known quantities gives

$$G_s = \frac{(10.0 - 6.38)\text{cm}^3}{(6.38 \text{ cm}^3)(0.60) - (10.0 \text{ cm}^3)(0.25)} = \mathbf{2.73} \quad (2.10.8)$$

EXAMPLE PROBLEM 2.11

General Remarks

This example problem illustrates the use of quantities derived from the Atterberg limits.

Problem Statement

A fine-grained soil has a liquid limit (LL) of 110% and a plastic limit (PL) of 56%. The clay content is 68% and the field moisture content of the soil is 60%.

- Compute the plasticity index (PI), the liquidity index (LI), and the activity (A).
- What is the predominant mineral in the soil?
- What is the soil state in the field?

Solution

- The PI is

$$PI = 110 - 56 = \mathbf{54\%} \quad (2.11.1)$$

The LI is

$$LI = \frac{w - PL}{PI} = \frac{60 - 56}{54} = \mathbf{0.074} \quad (2.11.2)$$

The activity is

$$A = \frac{PI}{\text{percent of clay} - \text{size fraction}} = \frac{54}{68} = \mathbf{0.79} \quad (2.11.3)$$

- b) Based on the activity, the predominant mineral is **illite**.
 c) Since the field moisture content is greater than the PL but is less than the LL, it follows that the soil is in a **plastic** state. This is also confirmed by the fact that the LI falls in the range $0 < LI < 1$.

EXAMPLE PROBLEM 2.12

General Remarks

This example problem illustrates some of the calculations involving the shrinkage limit.

Problem Statement

A saturated sample of clay has a field moisture content (w) of 32% and a shrinkage limit (SL) of 20%. What is the ratio of the dry volume (V_{dry}) of the sample to its original (natural) volume (V) if $G_s = 2.67$?

Solution

For the field conditions, from the general definition of the moisture content given in Eq. (1.14),

$$w = \frac{W_w}{W_s} \Rightarrow W_w = wW_s \quad (2.12.1)$$

The total volume of the saturated soil in the field is thus

$$V = V_w + V_s = \frac{W_w}{\gamma_w} + \frac{W_s}{G_s \gamma_w} \quad (2.12.2)$$

Since the weight of the solid phase remains unchanged in the SL test, combine the two above equations so as to eliminate W_w , giving

$$V = \frac{W_w}{\gamma_w} + \frac{W_s}{G_s \gamma_w} = \frac{W_s}{\gamma_w} \left(w + \frac{1}{G_s} \right) \quad (2.12.3)$$

At the SL,

$$w_{SL} = \frac{W_{wSL}}{W_s} \Rightarrow W_{wSL} = w_{SL} W_s \quad (2.12.4)$$

The total volume of the soil at the SL is thus

$$V_{dry} = V_{wSL} + V_s = \frac{W_{wSL}}{\gamma_w} + \frac{W_s}{G_s \gamma_w} = \frac{W_s}{\gamma_w} \left(w_{SL} + \frac{1}{G_s} \right) \quad (2.12.5)$$

The desired ratio of volumes is thus

$$\frac{V_{dry}}{V} = \frac{\frac{W_s}{\gamma_w} \left(w_{SL} + \frac{1}{G_s} \right)}{\frac{W_s}{\gamma_w} \left(w + \frac{1}{G_s} \right)} = \frac{w_{SL} + \frac{1}{G_s}}{w + \frac{1}{G_s}} \quad (2.12.6)$$

Substituting the given values of w , w_{SL} , and G_s gives the desired result; i.e.,

$$\frac{V_{dry}}{V} = \frac{w_{SL} + \frac{1}{G_s}}{w + \frac{1}{G_s}} = \frac{0.20 + \frac{1}{2.67}}{0.32 + \frac{1}{2.67}} = \mathbf{0.83} \quad (2.12.7)$$

Alternate Solution

It is also possible to solve this problem beginning with the following expression for the dry unit weight (recall Case 1.4 in Chapter 1)

$$\gamma_d = \frac{W_s}{V} = \frac{G_s \gamma_w}{1 + e} \Rightarrow W_s = \left(\frac{G_s \gamma_w}{1 + e} \right) V \quad (2.12.8)$$

For the field conditions,

$$W_s = \left(\frac{G_s \gamma_w}{1 + e} \right) V \quad (2.12.9)$$

At the SL,

$$W_s = \left(\frac{G_s \gamma_w}{1 + e_{SL}} \right) V_{dry} \quad (2.12.10)$$

Since the weight of the solid phase remains unchanged in the SL test, it follows that

$$\left(\frac{G_s \gamma_w}{1 + e} \right) V = \left(\frac{G_s \gamma_w}{1 + e_{SL}} \right) V_{dry} \Rightarrow \frac{V_{dry}}{V} = \frac{1 + e_{SL}}{1 + e} \quad (2.12.11)$$

Recalling Case 1.3 in Chapter 1, for saturated soil ($S = 100\%$), $e = wG_s$. Thus,

$$\frac{V_{\text{dry}}}{V} = \frac{1 + e_{\text{SL}}}{1 + e} = \frac{1 + w_{\text{SL}}G_s}{1 + wG_s} \quad (2.12.12)$$

Eq. (2.12.12) is seen to be equivalent to Eq. (2.12.7).

EXAMPLE PROBLEM 2.13

General Remarks

This example problem illustrates how a soil is classified according to the Unified Soil Classification System (USCS).

Problem Statement

A wet, dark brown soil that exudes an organic odor has 100% passing the No. 200 sieve. The natural liquid limit (LL) for the soil is 39%, the oven-dried liquid limit is 25%, and the plastic limit is 28%. Classify this soil according to the USCS.

Solution

Since 100% of the soil passes the No. 200 sieve, it is *fine grained*.

Since the LL is less than 50%, the soil will have *low plasticity*.

The plasticity index (PI) for the soil is $PI = 39 - 28 = 11\%$.

Figure Ex. 2.13 shows the location of the soil on Casagrande's plasticity chart.

The fact that the soil exudes an organic odor is consistent with the location of the soil on the plasticity chart in Figure Ex. 2.13, and indicates that the soil is organic.

Since

$$\frac{LL_{\text{oven-dried}}}{LL_{\text{natural}}} = \frac{25}{39} = 0.64 < 0.75 \quad (2.13.1)$$

the appropriate USCS group symbol will thus be **OL**. The associated group name will be *organic clay*.

EXAMPLE PROBLEM 2.14

General Remarks

This example problem illustrates how a soil is classified according to the Unified Soil Classification System (USCS).

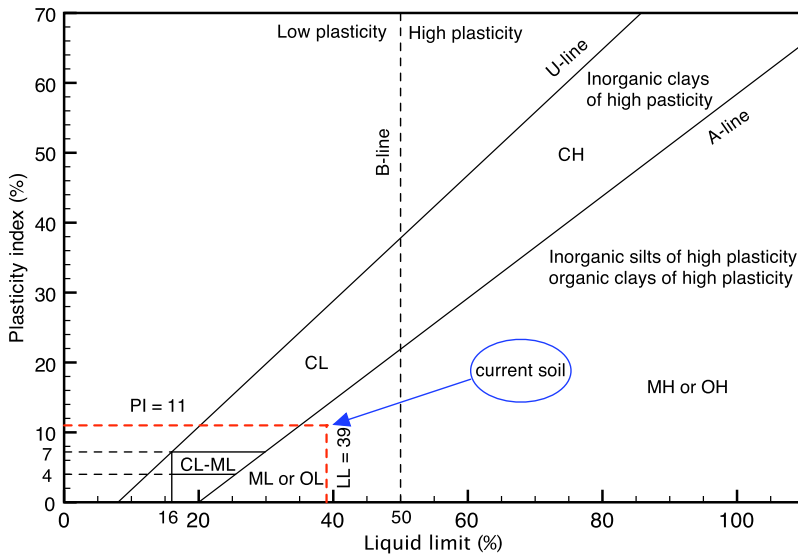


FIGURE EX. 2.13 Casagrande plasticity chart and location of current soil.

Problem Statement

A sample of gravel with sand has 74% fine to coarse subangular gravel, 24% coarse angular sand, and 2% fines. The maximum size of particles in the soil is 74 mm. The coefficient of curvature (C_c) is 2.7, and the coefficient of uniformity (C_u) is 12.6. Classify this soil according to the USCS.

Solution

Since the particle sizes in the soil do not exceed 75 mm, there are no cobbles present.

Since less than 5% fines are present, the soil will be a “clean gravel.”

Since $C_u > 4$ and $C_c < 3$, the appropriate group symbol will thus be **GW**.

The associated USCS group name will be a *well-graded gravel* (with sand).

EXAMPLE PROBLEM 2.15

General Remarks

This example problem illustrates how a soil is classified according to the Unified Soil Classification System (USCS).

Problem Statement

A sample of soil has 100% passing the No. 4 sieve and 75% passing the No. 200 sieve. The liquid limit (LL) for the soil was found to be 60%, while the plasticity index (PI) was 24%. Classify this soil according to the USCS.

Solution

Since more than 50% passes the No. 200 sieve, the soil is considered to be “fine grained”.

The point on the A-line corresponding to $LL = 60\%$ is

$$PI_{A-line} = 0.73(60 - 20) = 29.2\% \quad (2.15.1)$$

Since the soil's PI (24%) is less than 29.2%, it will plot *below* the A-line. As such, the appropriate group symbol will thus be **MH**; i.e., a high plasticity silt.

In addition, since 25% was retained on the No. 200 sieve, and since 0% was retained on the No. 4 sieve, the soil contains 25% sand.

The associated USCS group name will be *elastic silt with sand*.

EXAMPLE PROBLEM 2.16**General Remarks**

This example problem illustrates how a soil is classified according to the Unified Soil Classification System (USCS).

Problem Statement

A sample of soil has 60% passing the No. 4 sieve and 42% passing the No. 200 sieve. The liquid limit (LL) for the soil is 28%, and the plasticity index (PI) is 4.5%. Classify this soil according to the USCS.

Solution

Since less than 50% passes the No. 200 sieve, the soil is considered to be “coarse grained”.

Since $60\% > 50\%$ passes the No. 4 sieve, the soil will be a sand.

The point on the A-line corresponding to $LL = 28\%$ is

$$PI_{A-line} = 0.73(28 - 20) = 5.8\% \quad (2.16.1)$$

Since the soil's PI (4.5%) is less than 5.8%, it will plot *below* the A-line. Since the percent fines (42%) exceed 12%, the soil will contain clay.

The appropriate group symbol will thus be **SM-SC**; i.e., a *silty, clayey sand*.

Since 58% was retained on the No. 200 sieve, the “coarse” fraction is 58%. In addition, since 60% passed the No. 4 sieve, the gravel fraction = $100 - 60 = 40\%$. Finally, the sand fraction = $58 - 40 = 18\%$.

The associated USCS group name will be *silty, clayey sand with gravel*.

EXAMPLE PROBLEM 2.17**General Remarks**

This example problem illustrates how a soil is classified according to the Unified Soil Classification System (USCS).

Problem Statement

A sample of soil has 98% passing the No. 4 sieve and 74% passing the No. 200 sieve. The liquid limit (LL) for the soil is 58%, and the plasticity index (PI) is 30%. Classify this soil according to the USCS.

Solution

Since more than 50% passes the No. 200 sieve, the soil is considered to be “fine grained.”

The point on the A-line corresponding to $LL = 58\%$ is

$$PI_{A-line} = 0.73(58 - 20) = 27.7\% \quad (2.17.1)$$

Since the soil's PI (30%) exceeds 27.7%, it will plot *above* the A-line.

The appropriate group symbol will thus be **CH**; i.e., a *high plasticity clay*.

Since $100 - 74 = 26\%$ was retained on the No. 200 sieve, the “coarse” fraction is 26%. In addition, since 98% passed the No. 4 sieve, the gravel fraction = $100 - 98 = 2\%$. Finally, the sand fraction = $26 - 2 = 24\%$.

Since the percent gravel is less than 15%, the associated USCS group name will be *sandy fat clay*.